ME 445/545
Homework 4
Please submit to Gradescope

Submission guidelines: Must upload to the 445 or 545 gradescope pages. Must select corresponding pages, if not selected 5 points will be reduced.

Motivation: In our progression of developing the skills to be able to "design" a combustor we need a way to be able to calculate local temperatures, pressures, and species concentrations. These value are coupled, consequently we need a method to be able to simultaneously solve for the values. This need motivates deriving and understanding the governing equations. In this homework set you will gain more experience with using the governing equations (see problem 3) and in apply/using a special set of governing equations known as mixture fraction relationships. Solving for mixture fraction relationships allows us to avoid needing to solve for source terms in governing equations (i.e., reduce the number of equations that need to be solved).

1. (Turns 7.10) Consider a jet diffusion flame in which the fuel is methanol vapor (CH_3OH) and the oxidizer is air. The species existing within the flame are CH_3OH, CO, CO_2, O_2, H_2, H_2O, N_2, and OH.

   A. Determine the numerical value of the stoichiometric mixture fraction.
   B. Write out an expression for the mixture fraction, f, in terms of the flame-species mass fractions, Y_i. Assume all pairs of binary diffusion coefficients are equal, i.e., no differential diffusion. HINT: You will need to modify the approach used in the book because you cannot distinguish between the O which comes from the fuel or oxidizer.

2. (Turns 7.5) Consider the combustion of propane with air giving products, CO, CO_2, H_2O, H_2, and O_2, and N_2. Define the mixture fraction in terms of the various product species mole fractions, χ_i.

3. (Turns 7.2) Equation 7.55 expresses conservation of energy for a 1-D (Cartesian) reacting flow where no assumptions are made regarding the form of the species transport law or relationships between properties are not assumed. Starting with this equation, derive the Shvab-Zeldovich energy equation (Eq. 7.63) by applying Fick's law with effective binary diffusion and assuming $Le = 1$. Hint: Recall that $\dot{m}''' = \dot{m}''/dx$. You will need to determine how the product rule can be used to combine terms

4. GRAD: (Turns 7.6) The so-called "State relationships" are frequently employed in the analysis of diffusion flames. These state relationships related various mixture properties to the mixture fraction, or other appropriate conserved scalars. Construct state relationships for the adiabatic flame temperature Tad, and mixture density (ρ) for ideal combustion (i.e. no dissociation) or propane with air at 1 atm. Plot Tad and ρ as functions of the mixture fraction, f, for the range $0 \leq f \leq 0.12$. Hint: Think of this as if you were determining the temperature and density at different equivalence ratios. You can then relate mixture fraction to the equivalence ratio. Note that you

can assume ideal combustion products at fuel lean conditions. However, at fuel rich conditions both CO and CO_2 will be present. You will need to use the water-gas equilibrium reaction to have enough equations to solve for the concentrations of your products.

Javed Shahin | ME 445 | HW 4

Problem #1

- Consider a jet diffusion flame where (CH_3OH) Methanol vapor is the fuel and Air ($O_2 + 3.76N_2$)

Species in the flame:

$CH_3OH, CO, CO_2, O_2, H_2, H_2O, N_2,$ and OH

A) Determine - Stoichiometric mixture fraction

Stoich Chemical Rx: $CH_3OH + a(O_2 + 3.76N_2) \Rightarrow xCO_2 + yH_2O + zN_2$

C	1	$x = 1$
H	4	$2y = 4 \Rightarrow y = 2$
O	$1 + 2a$	$2x + y = 1 + 2a \rightarrow \frac{(4-1)}{2} = a = \frac{3}{2} = 1.5$
N	$3.76a$	$3.76a = z = 5.64$

Stoich: $CH_3OH + 1.5(O_2 + 3.76N_2) \Rightarrow CO_2 + 2H_2O + 5.64N_2$

$$f_{stoich} = Y_F = \frac{MW_{CH_3OH}\, X_{CH_3OH}}{MW_{mix}\, X_{mix}} = \frac{\text{Mass fuel}}{\text{Mass mix}}$$

		X moles
MW_{CH_3OH}	32.04 kg/Kmol	1
$MW\ O_2$	32 kg/Kmol	1.5
$MW\ N_2$	28.013 kg/Kmol	5.64

$$f_{stoich} = \frac{(32.04\ kg/Kmol)\ 1\ Kmol}{(32.04)1 + (32)1.5 + (28.013)(5.64)}$$

$= 0.1346$

B) Determine expression for mixture fraction (f)
- in terms of flame-species mass fraction Y_i

* Assume:
- All pairs of binary diffusion coefficients are equal
(ie. No differential diffusion)

$$f = \frac{\text{Mass of material in fuel stream}}{\text{Mass of mixture}} = \sum \frac{MW_{\text{fuel in } i}}{MW_{\text{of } i}} Y_i$$

Since Oxygen is present in the in both the oxidizer and in the fuel we can trace Carbon to get the mixture fraction.

mixture fracture that traces carbon

$$f_C = \left(\frac{MW_C}{MW_{CO_2}} Y_{CO_2} + \frac{MW_C}{MW_{CO}} Y_{CO} + \frac{MW_C}{MW_{CH_3OH}} Y_{CH_3OH} \right)$$

$$Y_i = \frac{M_i}{M_{tot}}$$

$$f_{CH_3OH} = \frac{MW_{CH_3OH}}{MW_C} f_C$$

$$= MW_{CH_3OH} \left(\frac{Y_{CO_2}}{MW_{CO_2}} + \frac{Y_{CO}}{MW_{CO}} + \frac{Y_{CH_3OH}}{MW_{CH_3OH}} \right)$$

Mixture fraction tracing Hydrogen

$$f_H = \left(\frac{MW_H}{MW_{H_2}} Y_{H_2} + \frac{MW_H}{MW_{H_2O}} Y_{H_2O} + \frac{MW_H}{MW_{OH}} Y_{OH} + \frac{MW_H}{MW_{CH_3OH}} Y_{CH_3OH} \right)$$

$$f_{CH_3OH} = \frac{MW_{CH_3OH}}{MW_H} f_H$$

$$= MW_{CH_3OH} \left(\frac{Y_{H_2}}{MW_{H_2}} + \frac{Y_{H_2O}}{MW_{H_2O}} + \frac{Y_{OH}}{MW_{OH}} + \frac{Y_{CH_3OH}}{MW_{CH_3OH}} \right)$$

Problem #2

- Consider the combustion of propane in Air giving the products
  * CO, CO_2, H_2O, H_2, + O_2, N_2
- Define:
  - Mixture fraction (f) in terms of product speicies mole fraction X_i

$$C_3H_8 + a(O_2 + 3.76N_2) \Rightarrow {}_uCO + {}_vCO_2 + {}_wH_2O + {}_xH_2 + {}_yO_2 + {}_zN_2$$

$$f = \frac{\text{Mass from fuel}}{\text{Total mass}} = \sum \frac{MW_{fi} \cdot mols_i}{MW_{mix} \cdot mols_{tot}} = \sum \frac{MW_{fi}}{MW_{mix}} X_i \;\Big|\; X_i = \frac{mol_i}{mol_{tot}}$$

$$f = \frac{MW_{C_3H_8}}{MW_{mix}} X_{C_3H_8} + \frac{MW_H}{MW_{mix}} X_{H_2O} + \frac{MW_H}{MW_{mix}} X_{H_2} + \frac{MW_C}{MW_{mix}} X_{CO} + \frac{MW_C}{MW_{mix}} X_{CO_2}$$

Problem #3

Equation 7.55 - expresses Conservation of energy 1-D reacting flow.

- in this regard No Assumptions where made involving the form of Species transport Law or relationships between properties.

* Starting w/ this equation — Derive - the Shvab-Zeldovich energy equation (Eq 7.63) Apply in Fick's Law w/ effective binary diffusion
  - Assuming: $Le = 1$
  - Recall: $\dot{m}''' = \frac{\dot{m}''}{dx}$, determine how the product rule can be used to combine like terms.

Conservation of energy 1-D reacting flow

$$\sum \dot{m}_i'' \frac{d h_i}{dx} + \frac{d}{dx}\left(-k \frac{dT}{dx}\right) + \dot{m}'' v_x \frac{d v_x}{dx} = \sum h_i \dot{m}_i'''$$

$\dot{m}_i''$ = mass flux ($\frac{kg}{s m^2}$)

$\dot{m}_i'''$ = mass production rate per unit volume ($kg/s m^3$)

V_x = Velocity in x

h_i = Species Enthalpy

T = Temp

Basic Assumptions: * S.S.
* No effect of gravity
* No shaft work or viscous dissipation, $W_s = 0$
* No radiation or heat transfer $\dot{Q} = 0$

Mass Transfer Laws: * Ordinary (Concentration ∇) diffusion only

* Constant Area

Applying species mass flux for Binary Diffusion (eq. 7.17)

$$\dot{m}_A'' = \dot{m}'' Y_A - \rho D_{AB} \nabla Y_A$$

* Note Ficks Law: $\dot{m}_A'' = Y_A(\dot{m}_A'' + \dot{m}_B'') - \rho D_{AB} \frac{dY_A}{dx}$

$$\dot{m}_{A,Diff}'' = -\rho D_{AB} \frac{dY_A}{dx} = -\rho D_{AB} \nabla Y_A$$

* $\dot{m}'' = \frac{\dot{m}}{A}$
* $\dot{Q}'' = \frac{\dot{Q}}{A}$

Examine Heat flux:

$$\dot{Q}''_x = -\frac{k\,dT}{dx} + \sum \dot{m}''_{i,diff}\, h_i$$

$$= -\frac{k\,dT}{dx} - \underbrace{\sum \rho D_{AB} \frac{dY_i\, h_i}{dx}}$$

$$\hookrightarrow \left(\sum h_i \frac{dY_i}{dx}\right) + \sum Y_i \frac{dh_i}{dx} = \sum \frac{dY_i h_i}{dx}$$

$$= -\frac{K\,dT}{dx} - \rho D_{AB} \sum h_i \frac{dY_i}{dx} + \rho D_{AB} \sum Y_i \frac{dh_i}{dx}$$

* $h = \sum h_i Y_i$ $\qquad \sum Y_i\left(\frac{dh_i}{dx}\right) = \sum Y_i\left(c_{p_i}\frac{dT}{dx}\right) = c_p \frac{dT}{dx}$

$$\Rightarrow \dot{Q}''_x = -\frac{K\,dT}{dx} - \rho D_{AB}\frac{dh}{dx} + \rho D_{AB} c_p \frac{dT}{dx}$$

* thermal Diffusivity $\alpha = K/\rho c_p \Rightarrow K = \alpha \rho c_p$

$$\dot{Q}''_x = \underbrace{-\alpha\rho c_p \frac{dT}{dx}}_{\text{flux of sensible enthalpy from conduction}} \underbrace{- \rho D_{AB}\frac{dH}{dx}}_{\text{flux of standardized enthalpy due to Binary species Diffusion}} + \underbrace{\rho D_{AB} c_p \frac{dT}{dx}}_{\text{flux of sensible enthalpy due to species diffusion}}$$

* Lewis # is Defined as $Le = \frac{\alpha}{D_{AB}} = 1 \Rightarrow \alpha = D_{AB}$

$$\therefore \dot{Q}''_x = -\rho D_{AB}\frac{dh}{dx}$$

Using the Energy Conservation w/ No work / No potential / No gravity

$$-\frac{d\dot{Q}''}{dx} = \dot{m}''\left(\frac{dh}{dx} + V_x \frac{dV_x}{dx}\right)$$

$$\Rightarrow \frac{d}{dx}\left(\rho D_{AB}\frac{dh}{dx}\right) = \dot{m}''\left(\frac{dh}{dx} + V_x\frac{dV_x}{dx}\right)$$

* $h = \sum Y_i h^\circ_{f,i} + \int_{T_{ref}}^{T} c_p\, dT$

$$\Rightarrow \frac{d}{dx}\left(\rho D_{AB}\sum h^\circ_{f,i}\frac{dY_i}{dx} + \rho D_{AB}\frac{d\int c_p dT}{dx}\right) = \dot{m}''\left(\sum h^\circ_{f,i}\frac{dY_i}{dx} + \frac{d\int c_p dT}{dx} + V_x\frac{dV_x}{dx}\right)$$

rearranging

$$\dot{m}'' \frac{d\int c_p\,dT}{dx} - \frac{d}{dx}\left(\rho D_{AB}\frac{d\int c_p\,dT}{dx}\right) + \dot{m}'' v_x \frac{dv_x}{dx} = -\frac{d}{dx}\left(\sum h^{\circ}_{f,i}\left(\dot{m}'' Y_i - \rho D_{AB}\frac{dY_i}{dx}\right)\right)$$

∵ Ficks Law $-\frac{d}{dx}\left(\sum h^{\circ}_{f,i}\underbrace{\left(\dot{m}'' Y_i - \rho D_{AB}\frac{dY_i}{dx}\right)}_{\dot{m}''_i}\right)$

$$\Rightarrow \quad -\frac{d}{dx}\left(\sum h^{\circ}_{f,i}\,\dot{m}''_i\right) = -\sum h^{\circ}_{f,i}\,\dot{m}'''_i$$

hence, The shvab-Zeldovich Energy Equation

$$\underbrace{\dot{m}'' \frac{d\int c_p\,dT}{dx}}_{\substack{\text{Rate of sensible}\\ \text{enthalpy transport by}\\ \text{convection (advection)}\\ \text{Per unit Volume } (W/m^3)}} - \underbrace{\frac{d}{dx}\left(-\rho D_{AB}\frac{d\int c_p\,dT}{dx}\right)}_{\substack{\text{Rate of Sensible}\\ \text{enthalpy transport}\\ \text{by Binary Diffussion}\\ \text{per unit volume } (W/m^3)}} + \overbrace{\dot{m}'' v_x \frac{dv_x}{dx}}^{\text{Kinetic Energy term}} = \underbrace{-\sum h^{\circ}_{f,i}\,\dot{m}'''_i}_{\substack{\text{Rate of sensible}\\ \text{enthalpy production}\\ \text{by Chemical Reaction}\\ \text{per unit volume } (W/m^3)}}$$

The kinetic energy term is generally small enough to be Neglected giving the general form of the equation

Shavb-Zeldovich Energy Equation:

$$\nabla\left(\dot{m}''\int c_p\,dT - \rho D_{AB}\nabla\left(\int c_p\,dT\right)\right) = -\sum h^{\circ}_{f,i}\,\dot{m}'''_i$$